1 Profile of Carbomer 940

1.1 Introduction

Carbomer 940 is a synthetic, high molecular weight polymer of acrylic acid, cross-linked with allyl ethers of pentaerythritol. It is an anionic rheology modifier used extensively to create crystal-clear gels and stabilize emulsions.

1.2 Physical and Chemical Properties

- **CAS Number:** 9003-01-4 or 76050-42-5
- **Appearance:** White, fluffy powder
- **Acidity (0.5% aq.):** pH 2.7 – 3.3
- **Viscosity (0.5% neutralized):** 40,000 – 60,000 mPa·s

1.3 The Grade Designation "940"

The "940" designation refers to the specific cross-linking density and molecular weight. It is characterized by *short rheology*, providing high yield value and high clarity. Compared to grades like 941 (long flow), 940 is designed for thick, non-flowing gels.

1.4 Mechanism of Gelation

1.4.1 Hydration in Water

When dispersed in deionized water, the polymer remains in a "tightly coiled" state due to internal hydrogen bonding. The equation for its weak dissociation is:

$$\text{R-COOH} + \text{H}_2\text{O} \rightleftharpoons \text{R-COO}^- + \text{H}_3\text{O}^+ \tag{1}$$

1.4.2 Neutralization with Triethanolamine (TEA)

Neutralization is the process of converting the acidic carboxyl groups into salts. Using Triethanolamine (TEA), the reaction is an acid-base neutralization:

$$\underbrace{\text{R-COOH}}_{\text{Polymer}} + \underbrace{\text{N}(\text{CH}_2\text{CH}_2\text{OH})_3}_{\text{TEA}} \rightarrow \underbrace{\text{R-COO}^-[\text{NH}(\text{CH}_2\text{CH}_2\text{OH})_3]^+}_{\text{Neutralized Salt}} \tag{2}$$

1.5 Why it Thickens

The addition of the base causes:

1. **Ionization:** The -COOH groups become negatively charged -COO$^-$.
2. **Electrostatic Repulsion:** The negative charges along the polymer chain repel each other.
3. **Uncoiling:** The polymer chain unfolds from a sphere into an extended structure.
4. **Solvation:** The extended 3D network traps water, increasing the viscosity of the medium exponentially.

2 Detailed Technical Analysis

2.1 What is Acrylic Acid?

Acrylic acid (CH_2=CHCOOH) is an organic compound and the simplest unsaturated carboxylic acid. It consists of a vinyl group connected directly to a carboxylic acid terminus. In Carbomer 940, thousands of these molecules are linked together to form the polymer backbone.

2.2 Structure of Allyl Ethers of Pentaerythritol

This component acts as the cross-linking agent. Pentaerythritol has a central carbon atom bonded to four $-CH_2OH$ groups. When converted to allyl ethers, some of these hydroxyl groups are replaced by allyl groups ($-CH_2-CH=CH_2$). These "arms" link different polyacrylic acid chains together, creating the 3D network required for gelation.

2.3 What is a Rheology Modifier?

A rheology modifier is an additive used to control the flow, viscosity, and texture of a formulation. Carbomer 940 is a "high-efficiency" modifier, meaning a very small amount (often more than 1%) can transform a thin liquid into a thick, stable gel.

2.4 Understanding the CAS Number

The CAS Registry Number is a unique universal identifier for chemical substances. For Carbomer 940, the CAS number is **9003-01-4** (or **76050-42-5**). This ensures that formulators and safety regulators are identifying the exact chemical structure regardless of the brand name used.

2.5 Non-flowing vs. Flowing Gels

- **Non-flowing (Short Flow):** Carbomer 940 produces a "short" rheology. This means the gel has high *yield value*—it will not flow under its own weight and maintains sharp peaks.
- **Flowing (Long Flow):** Other grades (like 941) produce "long" rheology, where the gel flows more easily and behaves more like a very thick liquid or syrup.

2.6 Uncoiling and Solvation Timing

1. **Uncoiling:** This occurs during the **Neutralization Phase**. As TEA is added, the H^+ ions are removed, leaving negative COO^- sites. The electrostatic repulsion between these negative charges forces the "coiled" polymer to snap open (uncoil).
2. **Solvation:** This happens immediately as the polymer uncoils. The exposed polar sites form hydrogen bonds with water molecules. The water becomes "trapped" within the expanded network, which is the physical process of thickening.

3 Physical States and Chemical Mechanism

3.1 Physical State Transition

The behavior of Carbomer 940 changes significantly depending on its environment. As seen in Figure 1, in its dry state, the polymer is tightly packed. Once added to water, it begins to hydrate and form an acidic dispersion.

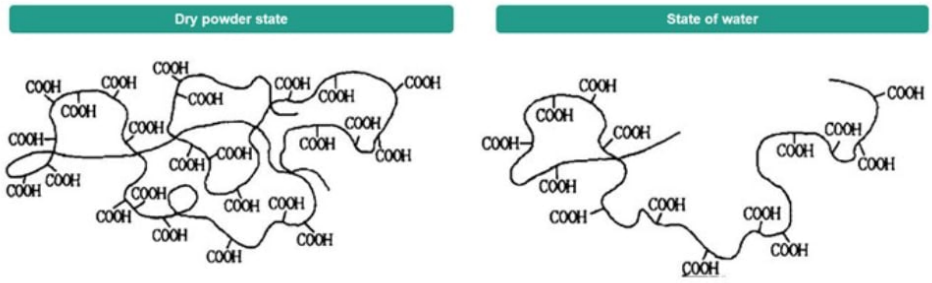

Figure 1: Molecular arrangement of Carbomer in dry powder state vs. water dispersion.

3.2 Neutralization and Application

The most critical phase in creating a gel is the addition of a base (like TEA). This process is visualized in Figure 2, showing how the coiled polymer "relaxes" into an uncoiled state, creating the thickening effect required for products like hand sanitizers.

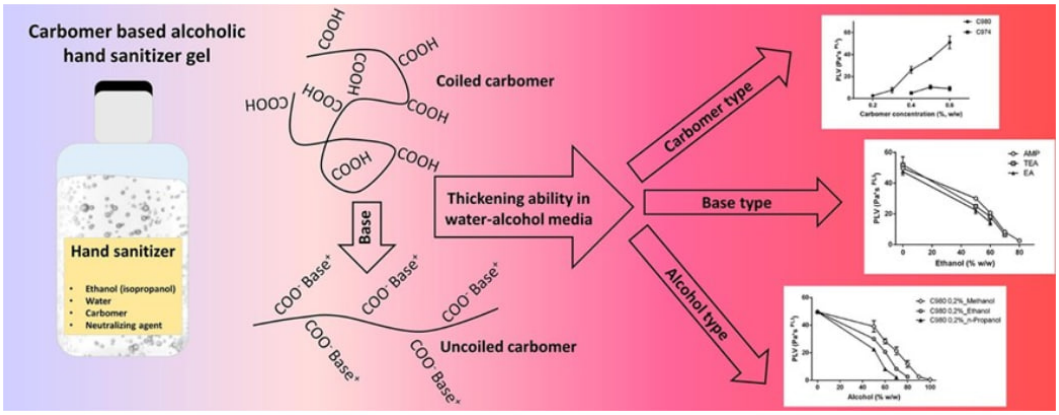

Figure 2: The mechanism of neutralization: transition from coiled to uncoiled state and its application in hand sanitizer formulations.

4 The 4-Step Thickening Mechanism

Based on the visual data in Figure 2, we can define the transformation in four distinct scientific steps:

1. **Ionization:** The COOH groups on the "Coiled carbomer" react with the **Base** to become negatively charged COO^- sites.
2. **Electrostatic Repulsion:** The negative charges now lining the chain push against one another.
3. **Uncoiling:** The polymer "explodes" outward from a ball into a long, straight chain (the **Uncoiled carbomer**).
4. **Solvation:** This expanded network traps water and alcohol molecules, which leads to the high-viscosity gel state seen in the **Hand Sanitizer** bottle.

5 Applications in Skin Care and Cosmetics

Carbomer resins are essential multi-purpose materials in the pharmaceutical and cosmetic industries. Their versatility allows for the creation of a wide range of product textures and stabilities.

5.1 Main Functions

In topical formulations, Carbomer 940 and related grades serve three primary roles:

- **Thickening:** Capable of producing a wide range of viscosities, from thin pourable liquids to thick, stiff gels.
- **Suspension:** It provides a high yield value, allowing it to permanently suspend insoluble ingredients (like beads, exfoliants, or silicone oil) in a formulation.
- **Emulsification:** It acts as a primary stabilizer, preventing the separation of oil and water phases in creams and lotions.

5.2 Key Advantages of Carbomer 940

The widespread use of Carbomer resins for over 50 years is due to the following specific benefits:

1. **Safety and Low Irritancy:** It is non-irritating to human skin and eyes. Toxicological studies have shown it to be exceptionally safe for long-term topical use.
2. **Superior Suspension Stability:** Compared to natural gums like cellulose or guar gum, carbomers provide much higher long-lasting suspension stability for ingredients like silicone oil.
3. **Foam Enhancement:** It improves foam stability in low-foam systems, contributing to a better sensory experience in cleansers.
4. **Electrolyte and Salt Resistance:** Specific cross-linked copolymers (such as Acrylate/C10-30 Alkyl Acrylate) maintain high viscosity even in systems containing salts or electrolytes.
5. **Efficiency in Processing:** It features a short wetting time and is easy to incorporate into industrial batches, saving significant time and resources.

6. **Microbial Resistance:** Carbomer is not easily decomposed by microorganisms and possesses inherent antiseptic properties, making the final product easier to preserve over time.

7. **Advanced Rheological Control:** It allows formulators to adjust the *thixotropic* properties of a gel, ensuring the product is stable in the bottle but spreads easily when applied to the skin.